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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 43.6

$$\int \frac{9}{4x^3 - 3x + 1} dx$$

Probeer $\frac{1}{1, 2, 4}$, nulpunt bij $x = \frac{1}{2}$

$$\int \frac{9}{(2x-1)(2x^2+x-1)} dx$$

$$\Delta = 1^2 - 4 \cdot 2 \cdot (-1) = 9 \quad x_{1,2} = \frac{-1 \pm \sqrt{9}}{2 \cdot 2} \sim x_1 = \frac{1}{2}, \quad x_2 = -1$$

$$\int \frac{9}{(2x-1)^2(x+1)} dx = \frac{A}{2x-1} + \frac{B}{(2x-1)^2} + \frac{C}{x+1}$$

Dan is

$$B = \frac{9}{x+1} \Big|_{x=\frac{1}{2}} = 6$$

$$C = \frac{9}{(2x-1)^2} \Big|_{x=-1} = 1$$

$$\begin{cases} \frac{9}{(2x-1)^2(x+1)} - \frac{B}{(2x-1)^2} = \frac{9}{(2x-1)^2(x+1)} - \frac{6}{(2x-1)^2} \\ = \frac{9 - 6(x+1)}{(2x-1)^2(x+1)} = \frac{-6x+3}{(2x-1)^2(x+1)} \\ = \frac{-3(2x-1)}{(2x-1)^2(x+1)} = \frac{-3}{(2x-1)(x+1)} \end{cases}$$

$$A = \frac{-3}{x+1} \Big|_{x=\frac{1}{2}} = -2$$

$$\Rightarrow \int \left(\frac{-2}{2x-1} + \frac{6}{(2x-1)^2} + \frac{1}{x+1} \right) dx$$

$$= \int \frac{-2}{2x-1} dx + \int \frac{6}{(2x-1)^2} dx + \int \frac{1}{x+1} dx$$

$$t = 2x - 1 \left(dt = 2dx, dx = \frac{dt}{2} \right)$$

$$= \int \frac{-2}{2x-1} dx + 6 \int t^{-2} \frac{dt}{2} + \int \frac{1}{x+1} dx$$

$$= \int \frac{-2}{2x-1} dx + 3 \int t^{-2} dt + \int \frac{1}{x+1} dx$$

$$= -\ln|2x-1| - \frac{3}{2x-1} + \ln|x+1| + C$$

References